

Course Code : CHT3001

UVXV/MW – 24/1528

**Third Semester B. Tech. (Electronics And Communication Engineering /
Electrical Engineering / Mechanical Engineering) Examination**

ENVIRONMENTAL SCIENCE

Time : 2 Hours]

[Max. Marks : 50

Instructions to Candidates :—

- (1) All questions are compulsory.
- (2) All questions carry marks as indicated against them.
- (3) Assume suitable data and illustrate answers with neat sketches wherever necessary.
- (4) Write chemical reaction wherever necessary.

1. (A) Answer the following questions (Any **Three**) :

- (i) Give the classification of air pollutants on the basis of their origin, composition and physical state ?
- (ii) What is National Ambient Air quality ? What are the different air pollutants considered during Air Quality Index ?
- (iii) What are different kinds of health effects occurring due to water pollutants ?
- (iv) What are the different steps involved in the treatment of waste water from dairy industry ? Explain any two in brief.

9(CO1)

(B) Choose the correct option from the following :

- (1) Which one of the following is not caused by polluted water?
 - (a) Cholera
 - (b) Dysentery.
 - (c) Typhoid
 - (d) Tuberculosis

- (2) The standard coagulant in water treatment is ?
(a) Hypochloride
(b) Chlorine
(c) Aluminium
(d) Alum
- (3) Which of the following is a secondary air pollutant ?
(a) SPM
(b) PAN
(c) SO₂
(d) NO₂
- (4) Which of the following causes Minamata disease ?
(a) Lead
(b) Mercury
(c) Magnesium
(d) Methyl chloride
- 4(CO1)

2. (A) Answer the following question (Any **Three**) :

- (i) What are the sources of hazardous solid wastes and their impact on human health ?
- (ii) Draw well labelled block diagram of four phase model of E waste Life Cycle of electricals and electronics equipment.
- (iii) What are the different steps/levels involved in the treatment in the E waste management ? Explain any one in detail.
- (iv) Given two machines producing 80 dB each, what is the total SPL ?
- 9(CO2)

(B) Choose the correct option from the following :

- (1) A 30 db increase in noise pollution levels represents :
(i) 30 db- fold increase in sound intensity.

- (ii) 100-fold increase in sound intensity
 - (iii) 1000-fold increase in sound intensity
 - (iv) 15-fold increase in sound intensity
- (2) The process of burning municipal solid wastes under suitable temperature and conditions in a specific furnace is called——

- (i) Landfill.
 - (ii) Incineration
 - (iii) Recycling
 - (iv) Vermi-composting
- (3) Under which rule of Government, guidelines for solid waste management are followed today ?
- (i) Municipal Solid Waste Rules, 2000
 - (ii) Municipal Solid Waste Rules, 2016
 - (iii) Solid Waste Rules, 2000
 - (iv) Solid Waste Rules, 2016
- 3(CO2)

3. (A) Answer the following (Any **Three**) :

- (i) What are the 12 principles of green chemistry ? Explain any two in brief.
 - (ii) What are the different advantages and disadvantages of green process ?
 - (iii) What are the different types of challenges to the sustainable development ? Explain any one in detail ?
 - (iv) Explain the role of Manufacturer in green computing ?
- 9(CO3)

(B) Choose the correct option from the following :

- (1) What is the ECO-mark ?
- (i) A scheme for labelling Environment Friendly Consumer product.

- (ii) A scheme for labelling pollution free industrial Unit.
 - (iii) A cost-effective production Technique.
 - (iv) An international certification recognizing Eco Friendly building.
- (2) Green chemistry improves ————— of chemical manufacturers.
- (i) Competitiveness
 - (ii) Easiness of production
 - (iii) Services
 - (iv) Chemicals
- (3) The sustainable development Goals comprises of how many goals and targets ?
- (i) 17 goals and 132 targets
 - (ii) 17 goals and 169 targets
 - (iii) 15 goals and 123 targets
 - (iv) More than one of the above
- (4) Which one of the following is Not a sustainable Development Goal (SDG) target to achieved by 2030 ?
- (i) Space research
 - (ii) Quality education
 - (iii) Gender equality
 - (iv) Zero hunger
- 4(CO3)

4. (A) Answer the following (Any **Two**) :

- (i) How do Article 48A and Article 51A(g) in the Indian Constitution articles reflect the responsibilities of the State and citizens towards the environment ?
 - (ii) Briefly discuss any two key environmental legislation in India that pertain to the protection of forests or wildlife.
 - (iii) What is Environmental Risk Assessment, and how does it play a role in pollution control and management ?
- 8(CO4)

(B) Choose the correct option from the following :

- (1) Which of the following is a key component of an Environmental Management System (EMS) as outlined in ISO 14001 ?
 - (i) Product Quality Control.
 - (ii) Financial Reporting
 - (iii) Environmental Policy and Objectives
 - (iv) Customer Satisfaction Surveys
- (2) Which EIA tool best identifies action-effect impact relations ?
 - (i) Interaction matrices
 - (ii) SWOT
 - (iii) Networks
 - (iv) Checklists.
- (3) Which of the following acts primarily governs the conservation and management of forests in India ?
 - (i) The Water (Prevention and Control of Pollution) Act, 1974
 - (ii) The Wildlife Protection Act, 1972
 - (iii) The Indian Forest Act, 1927
 - (iv) The Air (Prevention and Control of Pollution) Act, 1981
- (4) The agency which is not certifying the ISO ————
 - (i) Bureau of Indian Standard (BIS)
 - (ii) Quality Assurance Services
 - (iii) National Thermal Power Cooperation (NTPC)
 - (iv) All India institute of Local self Government. 4(CO4)

